

Trusted Intelligence at Wafer Scale

A Cerebras Confidential Computing Architecture (C-CCA)
for confidential LLM inference, cryptographic protection,
and AI agents

CPU CVM

Confidential WSE

Attestation

KMS / HSM

*Proposal status: new architecture / research direction — not a claim that current WSE-3/CS-4
already exposes a complete TEE.*

September 2026



Turn “fastest AI” into “trusted AI.”

The proposal is a security architecture extension for the Cerebras wafer-scale platform.

01

AI is becoming infrastructure of action

Agentic workloads create a new security boundary: the model is now connected to data, APIs, wallets, and enterprise credentials.

02

Confidential VMs stop at the CPU boundary

A CVM protects host memory, but a high-assurance AI platform must also attest and isolate the accelerator execution domain.

03

Cerebras can define a new TEE primitive

Make the WSE, its SRAM/fabric, model path, and CPU CVM one composite, remotely attested confidential workload.

EXECUTIVE OUTCOME A platform security moat that competitors cannot reproduce in software alone.

Sources: Cerebras CS-4 announcement (Aug 18, 2026); Cerebras company / sovereign AI materials; Intel TDX documentation; NVIDIA attestation documentation.

The AI workload is moving from conversation to action.

Speed makes agents practical; confidentiality makes them deployable into high-value environments.

THE TRAJECTORY

- 1 Chat** Prompt → response human-in-the-loop
- 2 Reasoning** Longer outputs latency matters
- 3 Agents** Tool calls + APIs credentials appear
- 4 Autonomy** Act on behalf of users trust becomes a prerequisite

CEREBRAS IS ALREADY POSITIONED AT THE FRONTIER

CS-4 3 × WSE-3 Turbo per rack-scale system

up to 30× faster inference vs. GPU systems
up to 10× more throughput per watt than CS-3

Agentic AI

Hyperscale

Sovereign AI

Regulated data

Strategic implication

At this scale, security must be designed into the platform — not bolted onto the host.

Source: Cerebras, "Introducing Cerebras CS-4: The Fastest AI Just Got Faster" (Aug 18, 2026).

A confidential VM protects the CPU — but the AI workload is larger than the CPU.

The proposed architecture makes the accelerator part of the cryptographic trust boundary.

CURRENT / INTERIM MODEL



Public Cerebras materials emphasize WSE/CS-4 performance and system architecture; they do not publicly establish a WSE-level TEE with remote attestation + protected execution memory.

PROPOSAL

Treat CPU CVM + WSE + model path + policy as one composite confidential workload.

No claim that this exists today — this deck proposes the architecture.

Define the adversary before defining the TEE.

C-CCA assumes the infrastructure is potentially hostile while the hardware security boundary remains trusted.

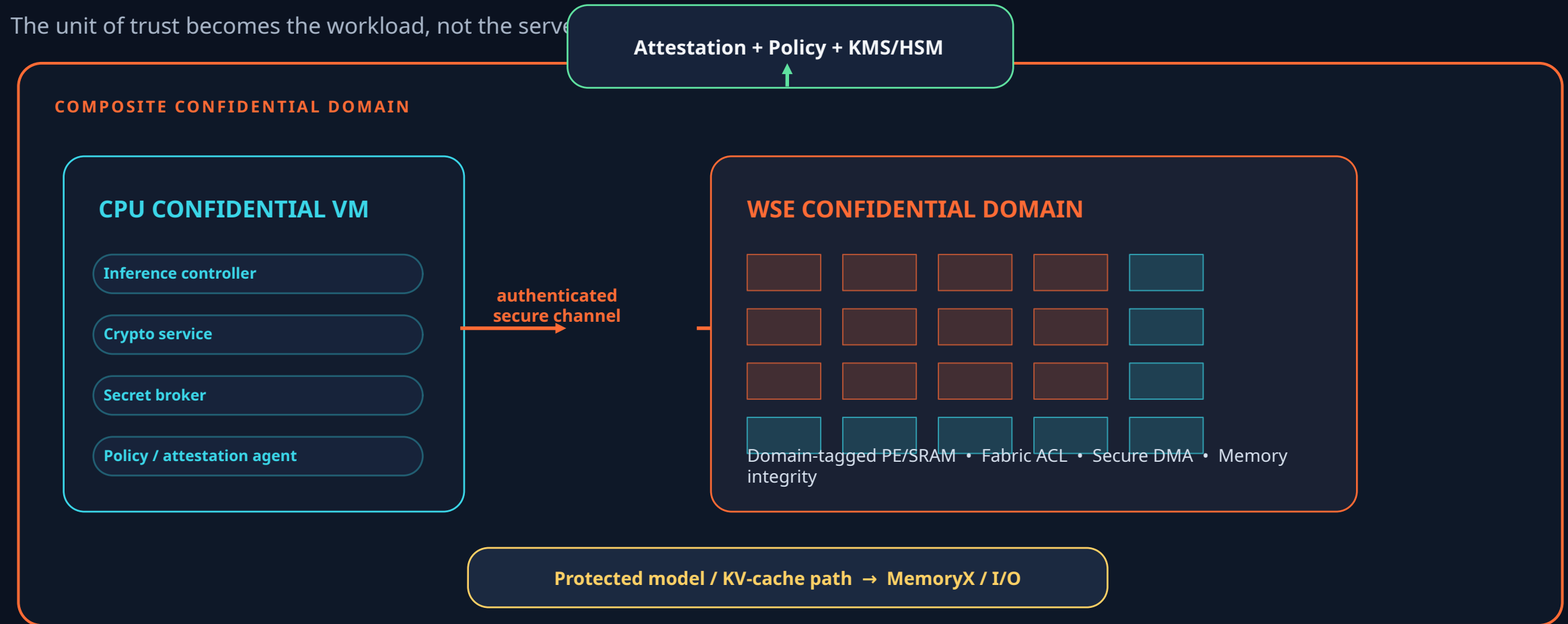
Adversary	Attack	C-CCA	Primary control
Cloud / data-center operator	Read host memory, inspect workload	YES	CVM + WSE isolation
Hypervisor / host OS	DMA, memory inspection, control-plane access	YES	Encrypted private memory + I/O controls
Unauthorized firmware / debug	Load altered firmware, inspect state	YES	Secure boot + measured boot + gated debug
Cross-tenant accelerator access	Read another domain's SRAM/fabric traffic	YES	Domain IDs + fabric ACL + protected DMA
Replay / rollback	Restore stale state / keys	YES	Counters / freshness + sealed state policy
Advanced physical side channels	Power / EM / invasive probing	PARTIAL*	High-assurance mode / packaging research

* Side-channel resistance should be an explicit tier, not an implicit promise. Confidentiality against privileged software is the first product target.

Conceptual threat model. Rooted in standard confidential-computing properties: memory confidentiality/integrity, hardware isolation, and remote attestation.

Composite TEE: CPU CVM + Confidential WSE + protected model path.

The unit of trust becomes the workload, not the server



Reference points: Intel TDX protected VM + attestation; NVIDIA GPU confidential computing + attestation; Cerebras WSE/CS-4 wafer-scale architecture.

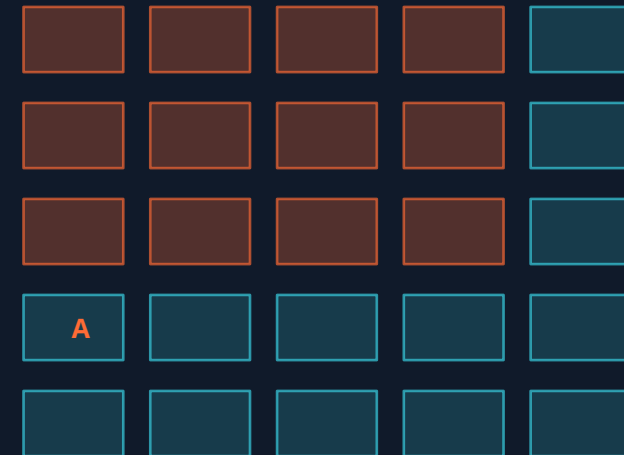
Apply enclave semantics to wafer-scale execution.

Use spatial security domains rather than attempting hundreds of thousands of tiny enclaves.

PROPOSED WSE SECURITY PRIMITIVES

- CDID** Confidential Domain ID on PE / SRAM / router / DMA
- MemEnc** Per-domain memory encryption + integrity metadata
- Fabric ACL** Domain-aware packet authorization + replay protection
- SecDMA** Only attested domains can issue protected transfers
- Secure Boot** Measured WSE firmware / runtime chain
- Debug Gate** Production debug access requires attested authorization

SPATIAL ISOLATION



B

Every memory region and fabric endpoint carries a security domain context.

Design goal A malicious host may control scheduling; it must not gain a valid capability to read or alter another domain's state.

Conceptual hardware design proposal. WSE-3/CS-4 architecture is from Cerebras public materials; the security mechanisms shown here are proposed.

Protect the model, prompt, KV cache, activations, and output.

Encryption at rest and TLS are not substitutes for confidentiality during execution.



WHAT STAYS CONFIDENTIAL

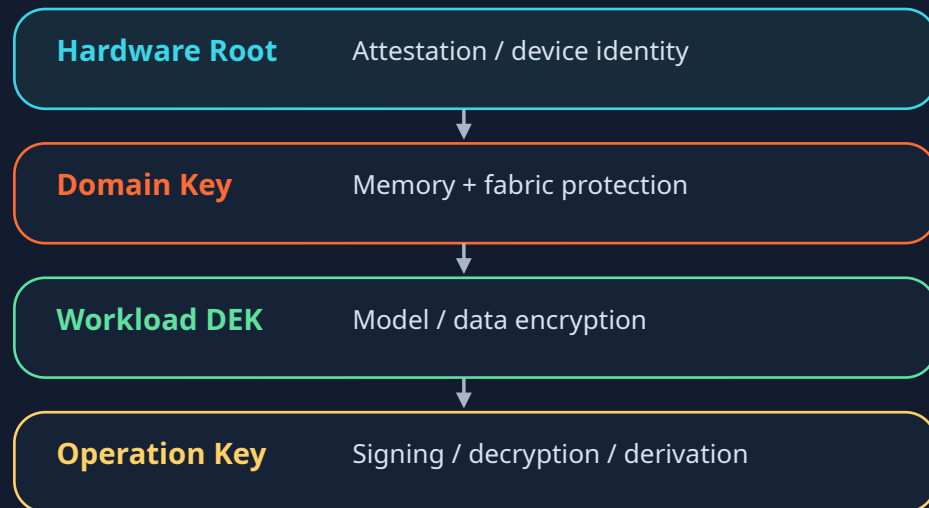


Cerebras: WSE-3 on-chip SRAM and high-bandwidth inference; NVIDIA/Intel: confidential computing uses hardware isolation + attestation to protect data-in-use.

Make plaintext private keys a last-resort resident of the TEE.

Keys should be released to attested code — not to the host, process environment, or operator.

KEY HIERARCHY



CRYPTO SERVICE: NON-EXPORTABLE HANDLES

- 1 Symmetric** AES-GCM • ChaCha20-Poly1305
decrypt() / encrypt()
- 2 Asymmetric** Ed25519 • ECDSA • BLS • PQC
sign() / derive()
- 3 Threshold** 2-of-3 / 3-of-5 shares
partial_sign() / aggregate()

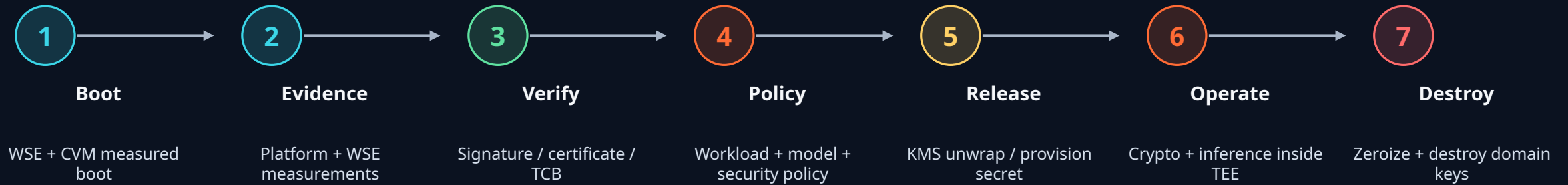
NO KEY EXPORT

Plaintext key lifetime = minimum needed for the cryptographic operation.

Design principle: KMS/HSM retains long-term authority; C-CCA is the last-resort execution location for plaintext key material.

Turn hardware trust into an enforceable policy decision.

The KMS should release secrets only after validating a composite evidence bundle.



COMPOSITE MEASUREMENT

$H(\text{CPU TEE} \parallel \text{WSE} \parallel \text{firmware} \parallel \text{runtime} \parallel \text{model} \parallel \text{policy})$

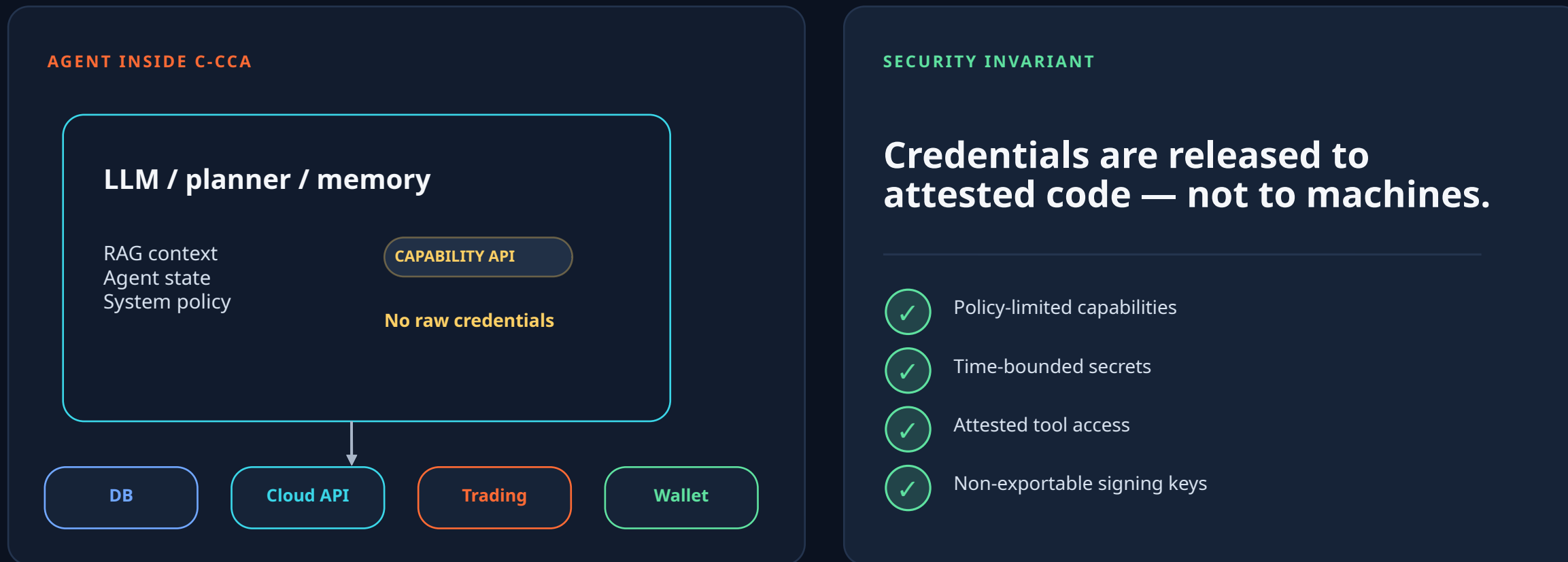
KEY-RELEASE CLAIM

"This exact workload may receive this exact secret."

Reference architecture: Intel TDX attestation; NVIDIA Attestation / H100 CC. Protocol choice can incorporate SPDm and EAT-style evidence.

Protect the LLM and the credentials it can use to act.

For agents, the highest-value secret may be a private key or API credential — not the prompt.



Cerebras CS-4 explicitly targets agentic workloads; C-CCA extends that trajectory with confidential credentials and attested tool access.

Where confidential Cerebras has the strongest economic pull.

Prioritize markets where data sensitivity + model IP + action authority overlap.



Cerebras already cites medical research, cryptography, energy and agentic AI among the domains using its systems.

Source: Cerebras company materials (current domains include medical research, cryptography, energy and agentic AI); sovereign AI materials.

Do not copy the GPU TEE. Create the wafer-scale TEE category.

The differentiator is the security architecture of the computing fabric itself.

Baseline CPU confidential VM	Accelerator CC GPU + TEE	C-CCA CPU + WSE composite	Outcome New category
VM memory confidentiality	✓	✓	✓
Accelerator attestation	—	✓	✓
Accelerator protected memory	—	✓	✓ *
Spatial execution-domain isolation	—	Limited	✓
Model / workload measurement	✓	✓	✓
Composite cluster attestation	—	✓	✓
MemoryX / wafer I/O protection	—	—	✓ *

** Proposed feature for next-generation WSE / secure-system design; not a statement of current product capability.*

Reference: Intel TDX; NVIDIA H100 confidential computing / attestation. Cerebras column is the proposed architecture.

Prototype the security plane before changing the silicon.

A staged path de-risks the proposal and produces an early customer story.



Key engineering gate

Demonstrate <2-5% end-to-end inference overhead target, measurable attestation latency, and zero plaintext secret exposure outside the protected domain.

Roadmap is a proposal. Exact performance and schedule targets should be validated by Cerebras architecture and security teams.

Security becomes a product differentiator, not a compliance checkbox.

The goal is to unlock workloads where performance alone is not enough.

01

Access

Sovereign, healthcare, finance, classified and high-value AI markets.

02

Monetize

Premium “Confidential AI” deployment tiers and managed services.

03

Defend

Protect Cerebras model/IP assets and reduce tenant/operator trust assumptions.

04

Differentiate

Define “trusted inference at wafer scale” as a category.

EXECUTIVE ASK

Authorize a joint C-CCA architecture sprint across Security + WSE/Hardware + Systems + Software, ending in a working attested-inference proof of concept and next-WSE requirements.

Success criterion: a customer can verify where its model, data, and cryptographic authority are — and prove that the cloud operator cannot access them.

Executive proposal. Commercial packaging and customer validation should follow the technical PoC.

Six rules should remain true regardless of implementation details.

These invariants can become the acceptance criteria for silicon, firmware, and software.

01 No secret release without valid attestation

02 No host access to confidential execution memory

03 No plaintext model leaves the trusted domain

04 No private-key export from the crypto service

05 No cross-domain fabric / DMA access

06 Destroying the domain destroys its operational key hierarchy

These are architecture invariants; implementations can evolve from software prototype to dedicated silicon.

Appendix — proposed acceptance criteria.